

# SEMIL PERIYASAMY

Applied AI/ML Engineer | LLMs · RAG Pipelines · Agentic Systems · Inference Optimization  
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## PROFESSIONAL SUMMARY

AI/ML Engineer with 2 years of experience designing and deploying production-grade agentic systems, end-to-end LLM pipelines, and high-throughput inference infrastructure. Proven track record fine-tuning large language models (Llama 3 70B), building scalable RAG and multi-agent workflows, and leveraging Gemini LLMs on GCP/Azure — directly aligned with Google's Applied AI/ML mandate to transform complex processes into AI-native solutions. Experienced with Python, SQL, PyTorch, and modern observability tooling (Weights & Biases).

## PROFESSIONAL EXPERIENCE

### Machine Learning Engineer | [AK Technologies](#)

Sep 2024 – Present | Coimbatore

- Architected and deployed production-ready multi-agent ERP assistant enabling autonomous document-driven queries and finance intelligence workflows mirrors Google's agentic systems mandate for self-sustaining, self-correcting agents.
- Fine-tuned Llama 3 70B on synthetic datasets; improved inference throughput by 40% via TensorRT-LLM, vLLM, INT8/FP8 quantization, and KV-cache optimization directly applicable to high-availability model endpoint design.
- Implemented distributed GPU infrastructure with Redis caching, reducing multi-node inference latency by 35%; designed health checks and fallback mechanisms for production reliability.
- Built scalable multilingual document intelligence system on GCP/Azure using Gemini 1.5 for automated translation, processing 1,000+ high-resolution requests/hour with full W&B observability for latency and model drift monitoring.
- Led RAG pipeline and document intelligence modules for geospatial urban transport planning platform end-to-end, from prototype to scaled production deployment.
- Designed Tamil-language Vision-OCR pipeline: structured extraction, LLM-powered translation, and document analytics experience in a regulated, high-accuracy, auditability-critical domain.

### AI Engineer Intern | [DigitalGarage Pvt. Ltd](#)

Mar 2024 – Aug 2024 | Coimbatore

- Developed and benchmarked ML models for computer vision and NLP applications using Python and TensorFlow; evaluated model performance across standard benchmarks.
- Supported full-stack AI-driven problem-solving initiatives, contributing to system design and model evaluation pipelines.

### Data Science Intern | [Shiash Info Solutions Pvt. Ltd](#)

Feb 2024 – May 2024 | Chennai

- Performed multi-source data analysis with Python (Pandas, NumPy) and built Seaborn/Matplotlib visualizations to communicate business insights to non-technical stakeholders.

## TECHNICAL SKILLS

|                |                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------|
| Languages:     | Python, SQL (PostgreSQL, Microsoft SQL Server), R                                                   |
| Generative AI: | LLM Fine-tuning (LoRA/QLoRA), RAG Pipelines, Agentic Systems, Synthetic Data Generation, Gemini 1.5 |
| Inference:     | TensorRT-LLM, vLLM, CUDA, Quantization (INT8/FP8), KV-Cache Optimization, High-Throughput Serving   |
| ML Frameworks: | PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy                                                    |
| Cloud & MLOps: | Google Vertex AI, Azure Machine Learning, Docker, Redis, Weights & Biases (W&B), GCP                |

## CERTIFICATIONS & OPEN-SOURCE

**NVIDIA DLI:** Exploring Adversarial Machine Learning (Certified) | **NVIDIA:** Deploying RAG Pipelines for Production at Scale | **IBM Professional AI Developer** (Coursera)

**Open Source:** Contributor to **vLLM** (KV-cache & high-throughput inference docs) and **LLMCache** (cache reuse strategies & memory efficiency for repeated prompt workloads).

## EDUCATION

### B.Tech – Artificial Intelligence & Data Science

Jul 2020 – May 2024

Arjun College of Technology, Coimbatore | GPA: 8.2/10